Zuo Gou

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EDUCATION

Tsinghua University Sep 2023 – Expected Jun 2028

B.Eng., Automation (Xinya College & Dept. of Automation) *Beijing, China*

· **GPA: 3.8/4.0** **Awards**: Tsinghua Comprehensive Scholarship (2024, 2025); Luo Yuehua Scholarship (2024)

PUBLICATIONS AND MANUSCRIPTS

[1] **Z. Gou***, C. Lu*. *Detecting Silent Faults at the Environment–Trainer Boundary in Reinforcement Learning.* CoRL 2026 Workshop on Everything Beneath the Policy. arXiv:2611.08144. (*equal contribution)

[2] **Z. Gou** et al. *Relative Noise Schedules in Joint Continuous–Discrete Diffusion.* In preparation; target ICML 2027.

[3] **Z. Gou** et al., W.-C. Ma. *Humanoid2Humanoid: Shared Sensing for Cooperative Loco-Manipulation.* In preparation; target RSS 2027.

RESEARCH EXPERIENCE

Prof. Wei-Chiu Ma's Group (Dept. of Computer Science, Cornell University) Jun 2026 – Nov 2026

Research Intern, Multi-Agent Humanoid Locomotion & Inter-Robot Communication *Ithaca, NY*

· Studying how two humanoids can share what they see in real time, so that a robot walking backward climbs stairs hidden from its own cameras by relying on its partner's view.

· Established when such sharing matters: robots usually infer hidden terrain from gait and contact, and only in specific conditions does the partner's view become indispensable, recovering 20–40 success points.

· Built decentralized policies that act on their own sensing plus messages from the partner, halt safely when the link drops and resume when it returns; demonstrated a simulated pair climbing stairs together.

Prof. Ge Liu's Group (Dept. of Computer Science, University of Illinois Urbana-Champaign) Mar 2026 – Present

Research Intern, Diffusion Models & Generative Modeling Theory *Remote*

· Studying how a diffusion model that generates continuous and discrete quantities at once — atom positions with atom types and bonds — should pace noise across its two channels, a choice now hand-tuned per dataset.

· Characterized when the likelihood bound is indifferent to that pacing and what the best schedule looks like when it is not, and confirmed the predictions on public molecular models.

· Showed that reported likelihoods are dominated by a single coarse step of the sampler, and built an estimator reaching the same precision with roughly 2,600× less computation.

Prof. Huazhe Xu's Group (IIIS, Tsinghua University) Jul 2025 – Jan 2026

Research Assistant, Vision–Language–Action Models & RL *Beijing, China*

· With Chenhao Lu, co-developed *PlugRL* (plugrl.github.io; open source, ongoing), a wire protocol that lets RL train environments on other machines, in other languages or outside a trainer's cluster; wrote its specification and conformance checkers.

· Showed that silent faults at the environment–trainer boundary mostly slip past learning-curve checks and clean evaluation, while a step ledger kept on both sides caught and located all 585 faulty runs it checked, for a median 3% of training time on Pendulum [1].

Prof. Yebin Liu's Group (Dept. of Automation, Tsinghua University) Sep 2024 – Jun 2025

Research Assistant, Dexterous Manipulation & Diffusion-based Control *Beijing, China*

· Developed a diffusion model that generates physically plausible hand–object interaction for dexterous manipulation, with the simulation environments and demonstration pipelines it was trained on.

SKILLS AND SERVICE

Programming	Python (PyTorch), C++, CUDA, MATLAB, Shell
Robotics & simulation	IsaacGym, Isaac Lab, MuJoCo, Legged Gym, RSL-RL
Infrastructure	Linux, Slurm clusters, distributed multi-GPU training, Weights & Biases
Service	Reviewer, CoRL 2026 Workshop on Everything Beneath the Policy